

DAQ Core

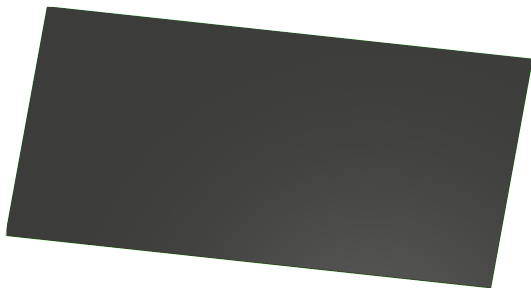
Variant: DRAFT

2026-09-29

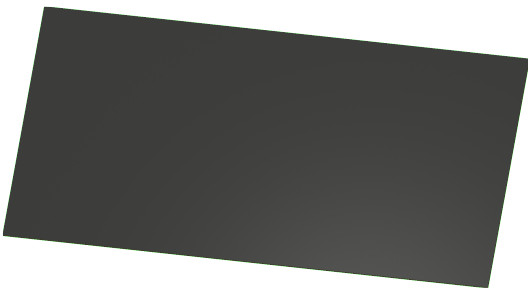
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	Power	21		31	
2	Block Diagram	12	Power - Sequencing	22		32	
3	Project Architecture	13	Revision History	23		33	
4	MCU	14		24		34	
5	MCU Expander	15		25		35	
6	MCU IOT	16		26		36	
7	In&Out - ADC	17		27		37	
8	In&Out - Command	18		28		38	
9	Serial Interface	19		29		39	
10	Sensor	20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 29-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

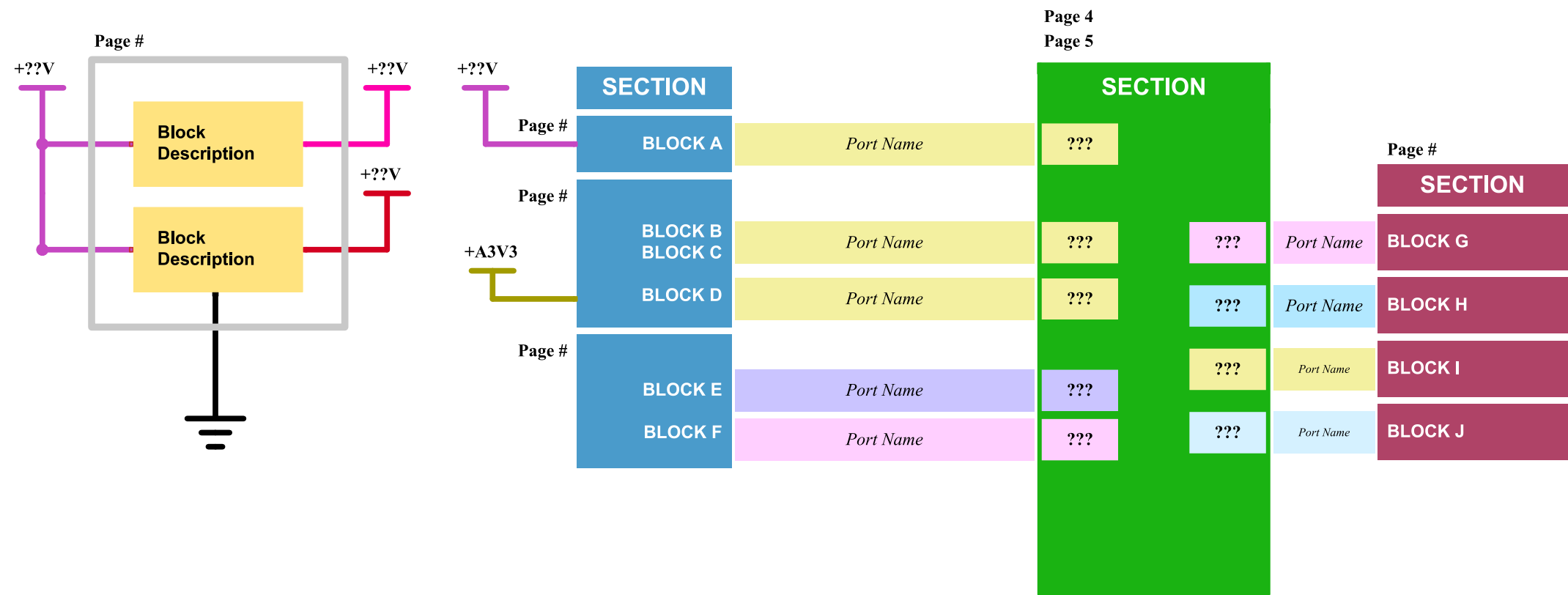
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:	Company: Nexvantis	Variant: DRAFT	Git Hash: 41b59e0
	Sheet Title:	Board Name: DAQ Core	Project Name: Data Acquisition Board	
	Sheet Path: /	File Name: Data-Acquisition-Board.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12
		Reviewer:	Size: A3	Revision: + (Unreleased)
			Sheet: 1 of 13	

[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
		Board Name: DAQ Core		Project Name: Data Acquisition Board	
	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/			Reviewer:	Size: A3	Sheet: 2 of 13

[3] Project Architecture

Sensor



Page 8

File: Sensor.kicad_sch

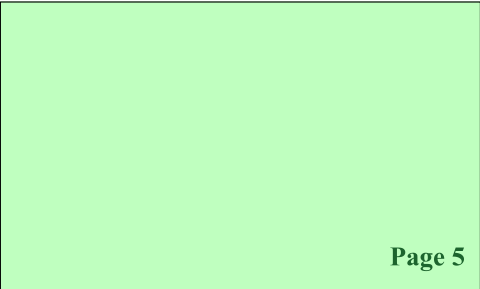
MCU



Page 4

File: MCU.kicad_sch

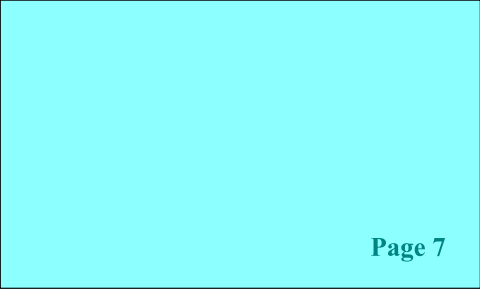
In&Out - ADC



Page 5

File: In&Out - ADC.kicad_sch

Serial Interface



Page 7

File: Serial Interface.kicad_sch

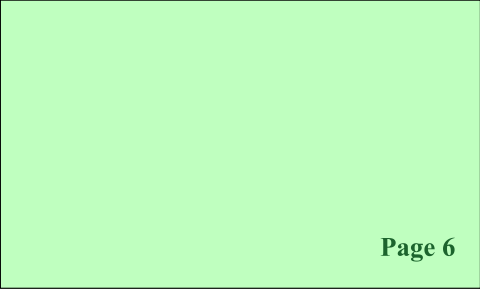
MCU Expander



Page 4

File: MCU Expander.kicad_sch

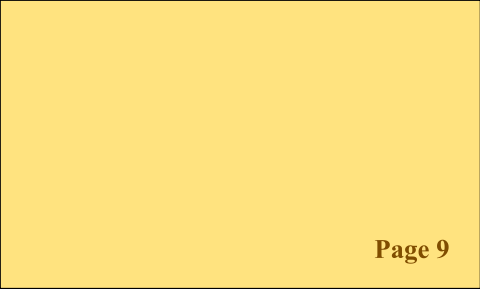
In&Out - Command



Page 6

File: In&Out - Command.kicad_sch

MCU IOT



Page 9

File: MCU IOT.kicad_sch

Power



Page 10

File: Power.kicad_sch

LAYOUT NOTE:
blablabla

DESIGN NOTE:
blablabla

DESIGN NOTE:
blabla

DESIGN NOTE:
blabla

LAYOUT NOTE:
blabla

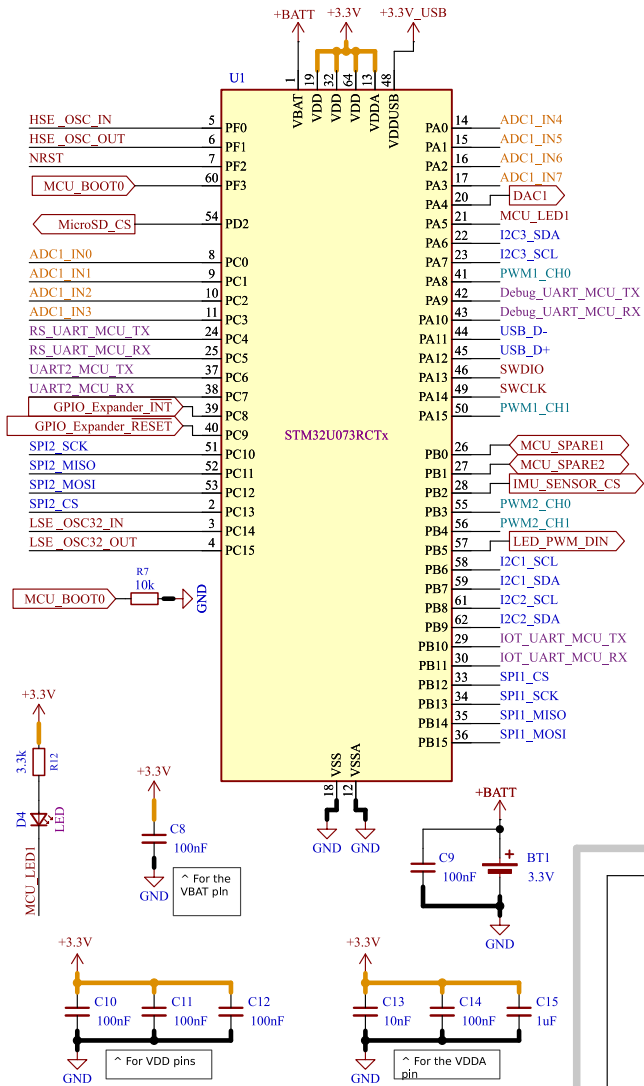
Block Description

Description

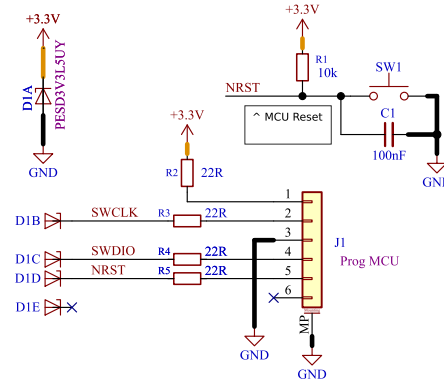
	Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Sheet Title: Project Architecture	Board Name: DAQ Core		Project Name: Data Acquisition Board	
	Sheet Path: /Project Architecture/	File Name: Project Architecture.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
	Reviewer:			Size: A3	Sheet: 3 of 13

[4] MCU

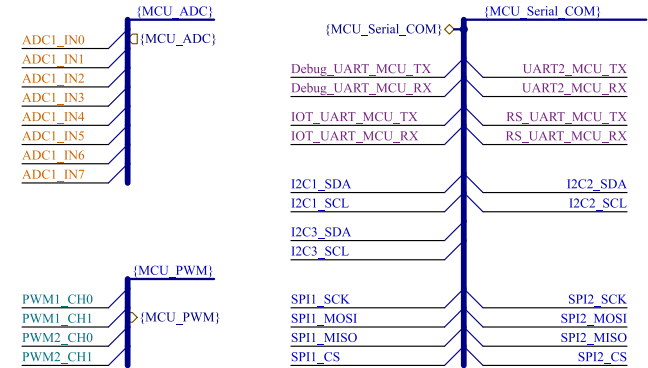
STM32U073RCTx



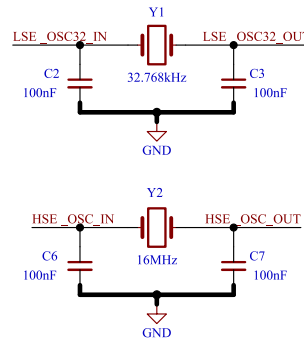
Programming



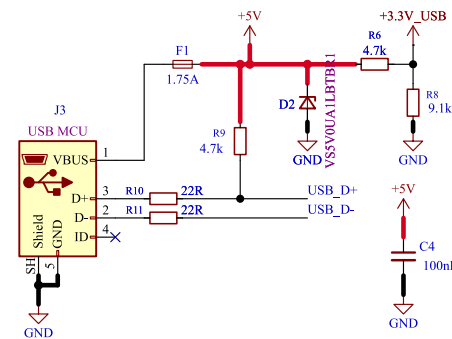
BUS



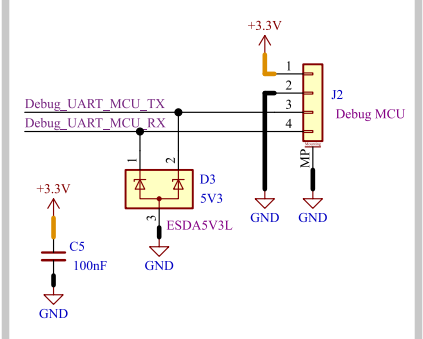
Quartz



USB



Debug



Comments:

Company:

Variant:

Git Hash:

Nexvantis

DRAFT

41b59e0

Board Name:

Project Name:

DAQ Core

Data Acquisition Board

Sheet Title:

File Name:

Designer:

Date:

Revision:

MCU

MCU.kicad_sch

Jordan Nzokou

2025-01-12

+

Sheet Path:

Reviewer:

Size:

Sheet:

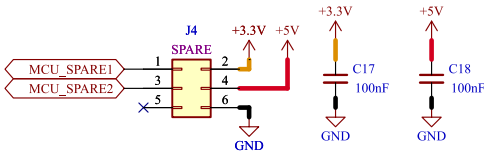
/Project Architecture/MCU/

A4

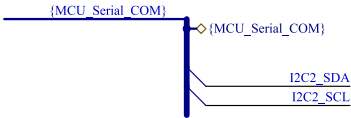
4 of 13

[5] MCU Expander

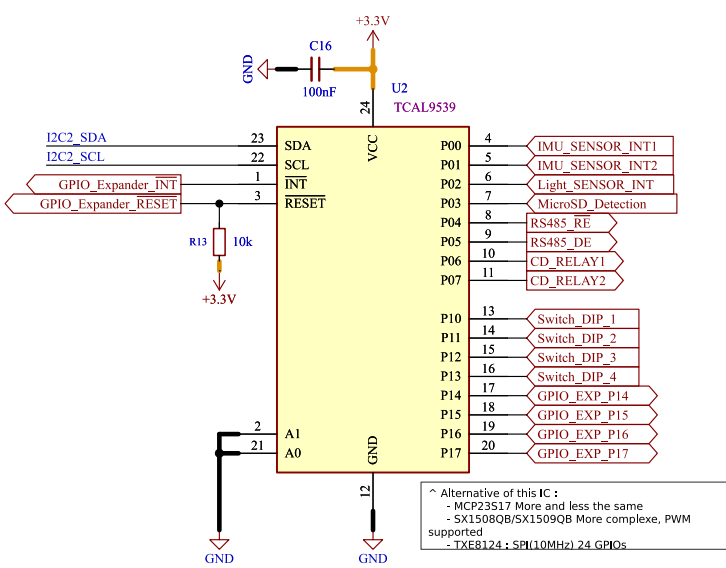
MCU SPARE CONNECTOR



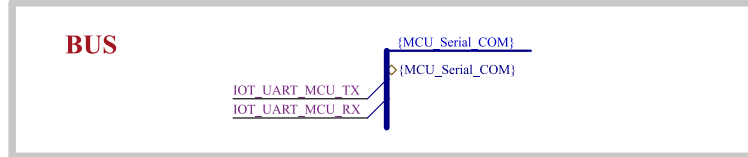
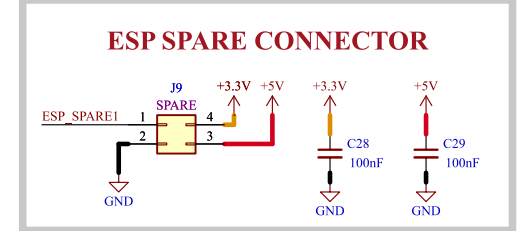
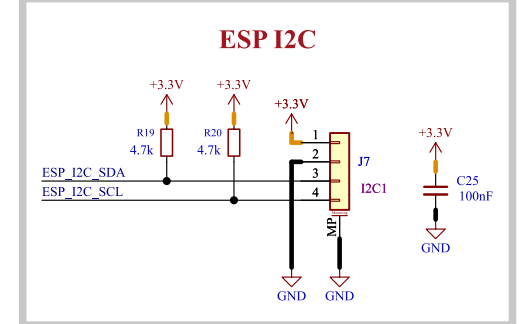
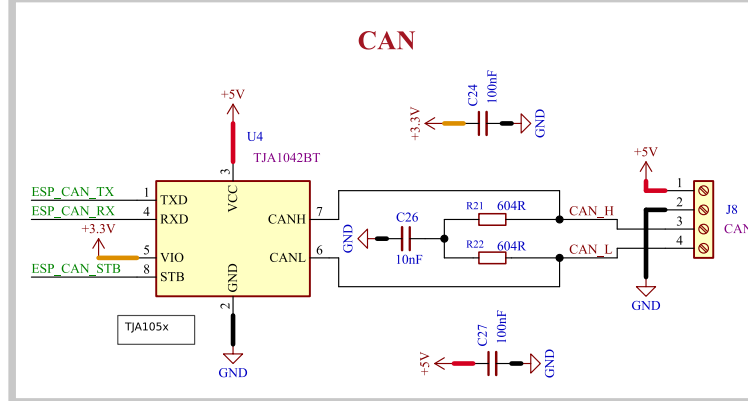
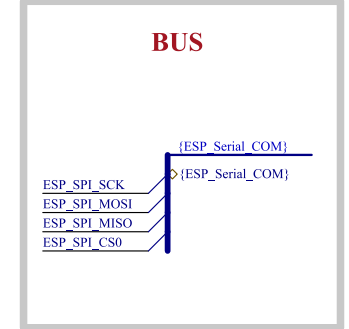
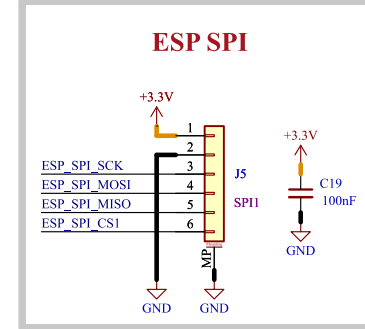
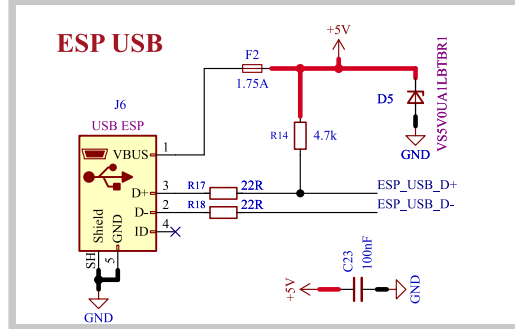
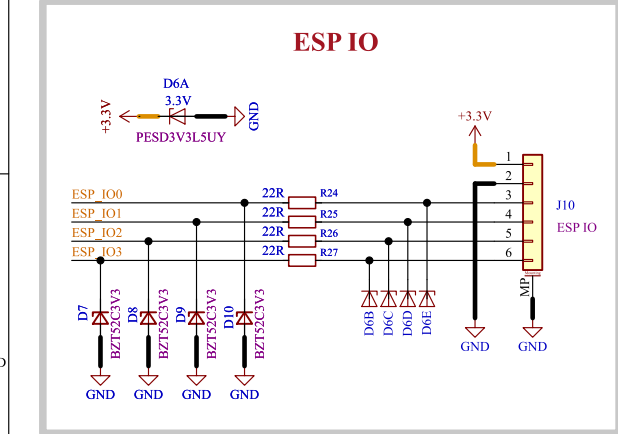
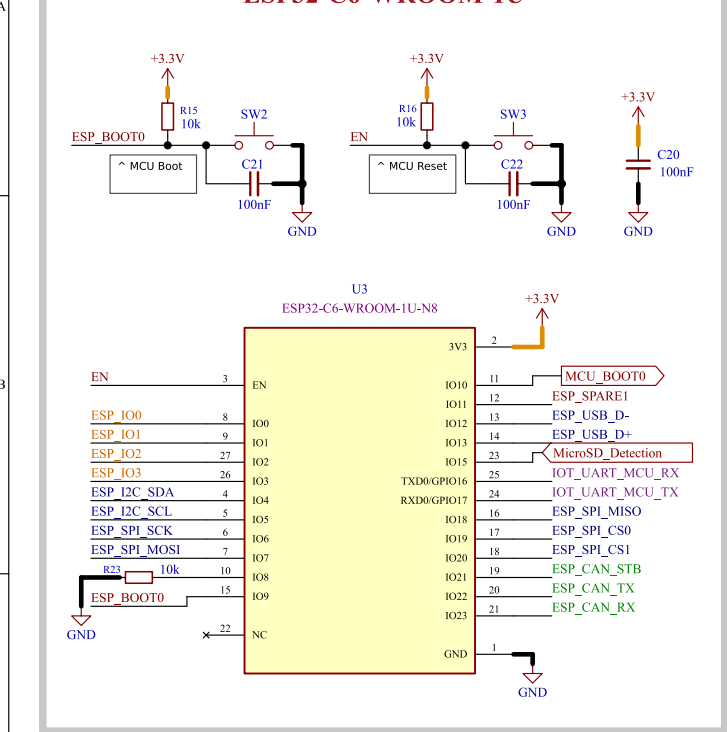
BUS



GPIO Expander



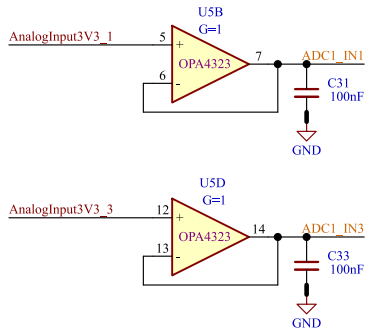
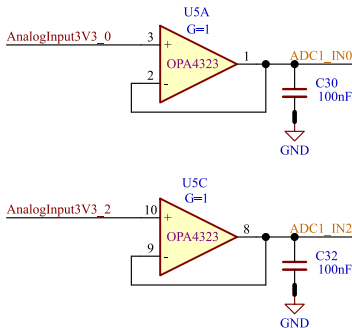
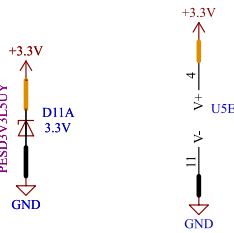
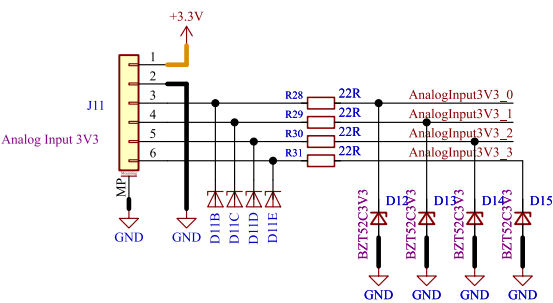
Comments:	Company:		Variant:	Git Hash:
	Nexvantis		DRAFT	41b59e0
	Board Name:		Project Name:	
Sheet Title:	File Name:	Designer:	Date:	Revision:
MCU Expander	MCU Expander.kicad_sch	Jordan Nzokou	2025-01-12	+(Unreleased)
Sheet Path:		Reviewer:	Size:	Sheet:
/Project Architecture/MCU Expander/			A4	5 of 13



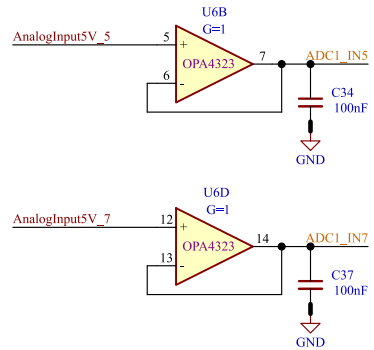
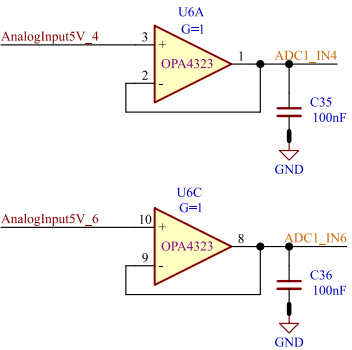
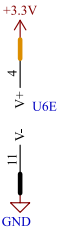
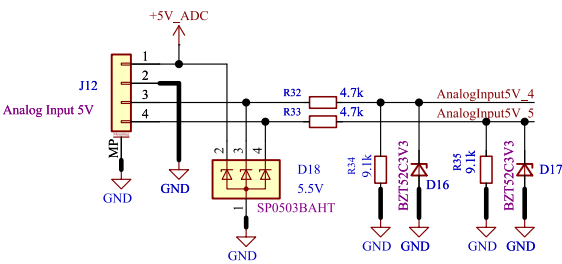
	Comments: ESP32-C6-WROOM-1U Datasheet TJA1042BT Datasheet		Company: Nexvantis		Variant: DRAFT		Git Hash: 41b59e0			
			Board Name: DAQ Core			Project Name: Data Acquisition Board				
	Sheet Title: MCU IOT		File Name: MCU IOT.kicad_sch		Designer: Jordan Nzokou		Date: 2025-01-12		Revision: + (Unreleased)	
	Sheet Path: /Project Architecture/MCU IOT/				Reviewer:		Size: A4		Sheet: 6 of 13	

[7] Input&Output ADC

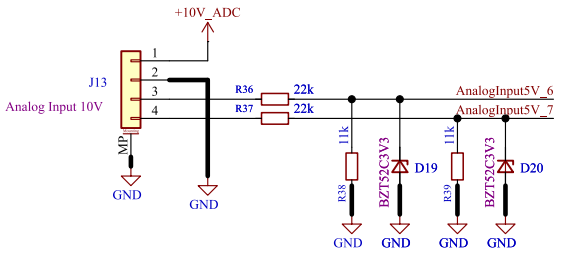
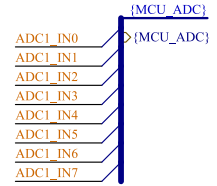
Analog Input ADC 3V3



Analog Input ADC 5V & 10V



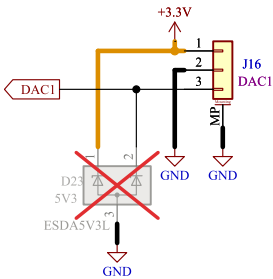
BUS



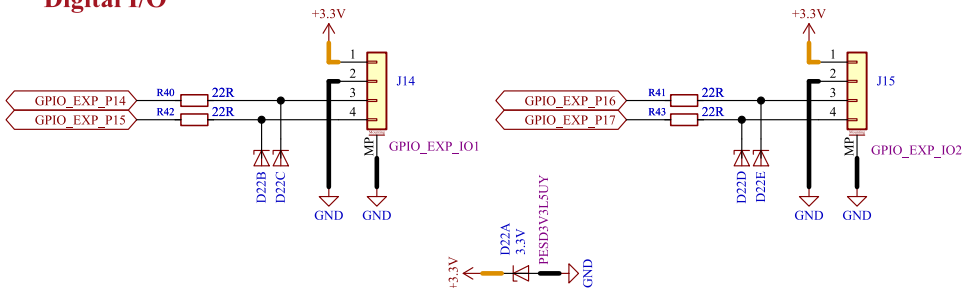
Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
	Board Name: DAQ Core		Project Name: Data Acquisition Board	
Sheet Title: Input&Output ADC	File Name: In&Out - ADC.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Project Architecture/In&Out - ADC/		Reviewer:	Size: A4	Sheet: 7 of 13

[8] Input&Output Command

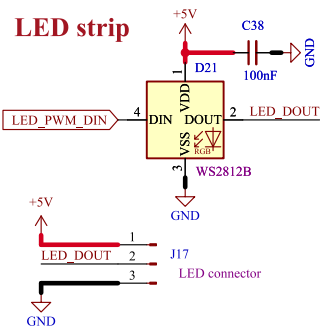
DAC



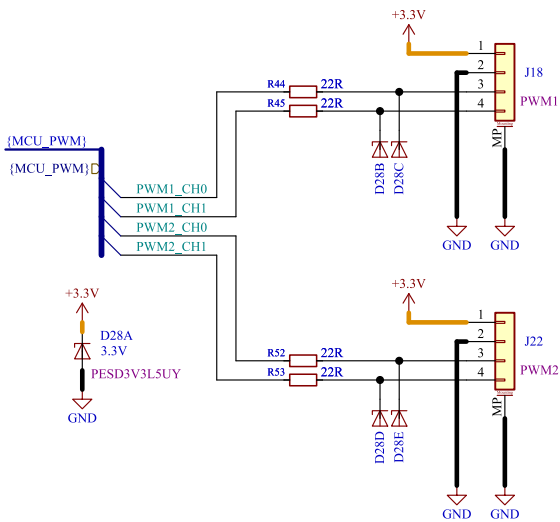
Digital I/O



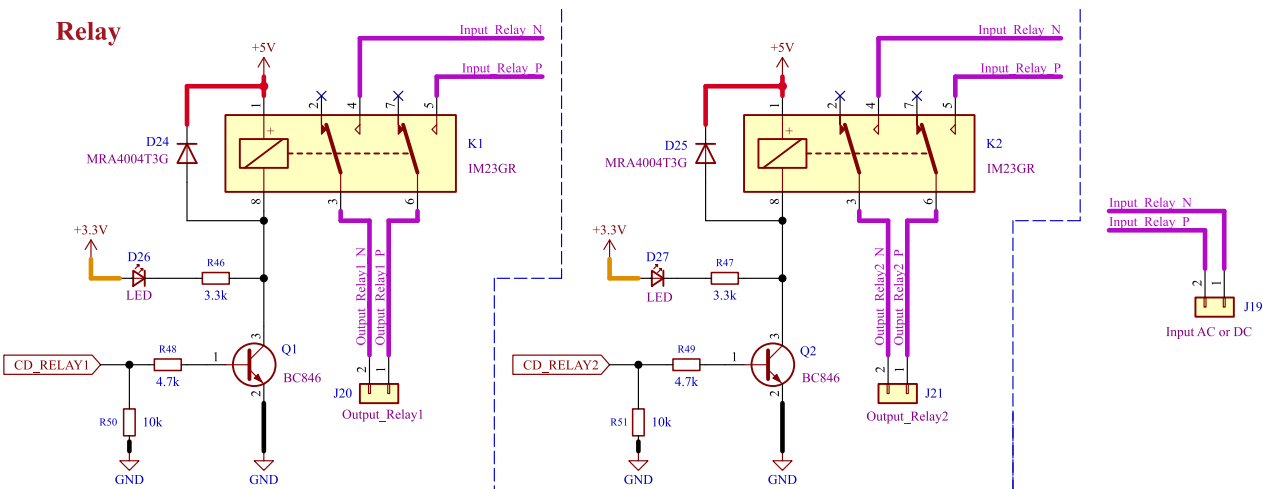
LED strip



PWM



Relay



Comments:

Company:

Variant:

Git Hash:

Nexvantis

DRAFT

41b59e0

Board Name:

Project Name:

DAQ Core

Data Acquisition Board

Sheet Title:

File Name:

Designer:

Date:

Revision:

Input&Output Command

In&Out - Command.kicad_sch

Jordan Nzokou

2025-01-12

+

Sheet Path:

Reviewer:

Size:

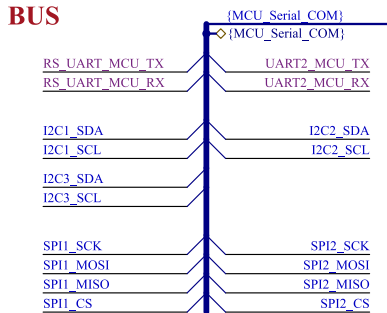
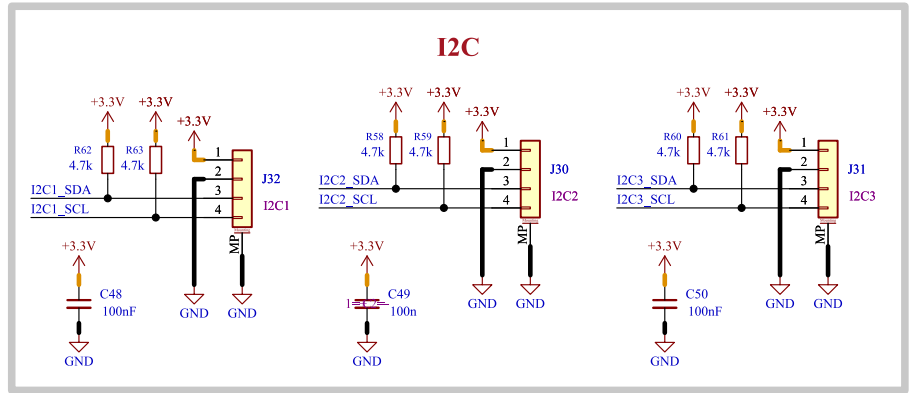
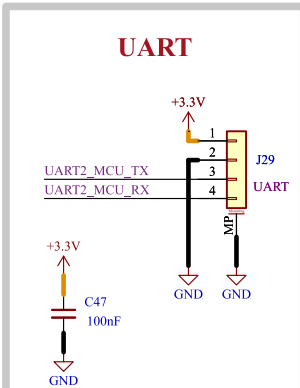
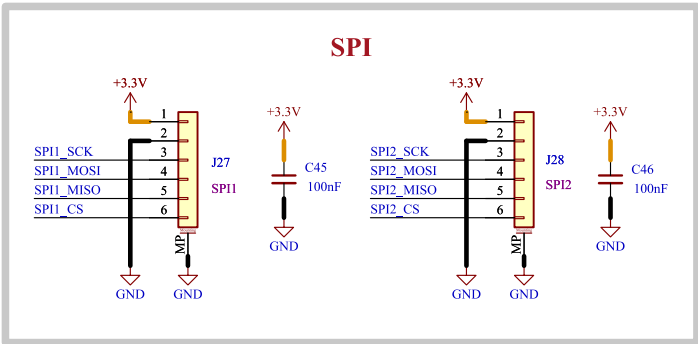
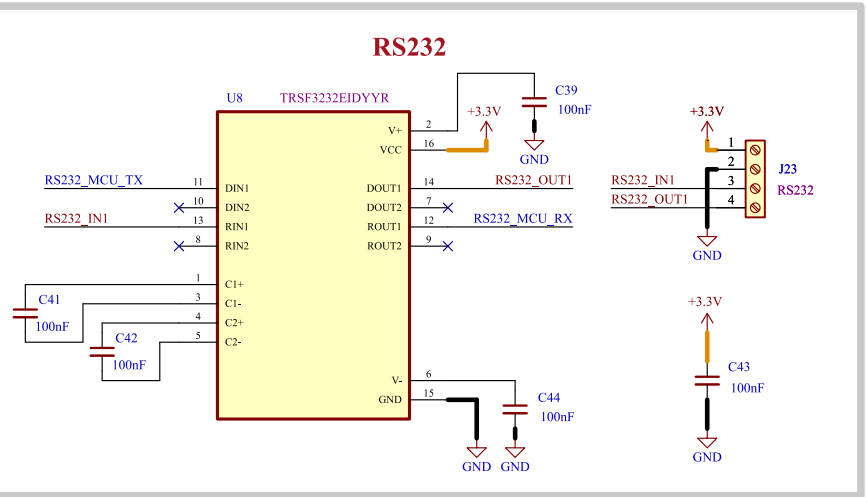
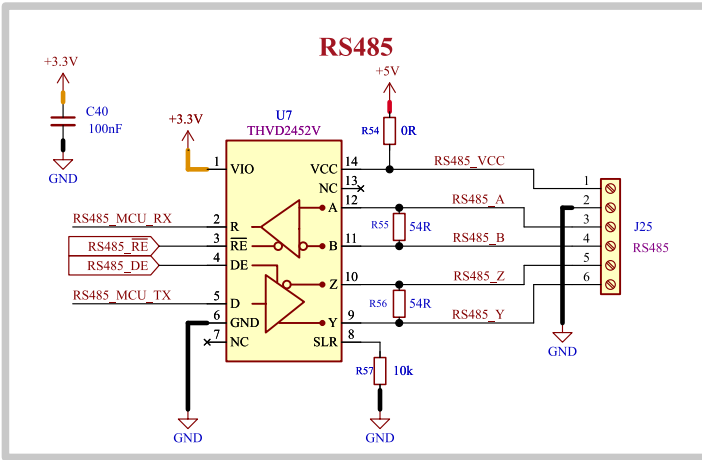
Sheet:

/Project Architecture/In&Out - Command/

A4

8 of 13

[9] Serial Interface

Comments:

Sheet Title:

Serial Interface

Sheet Path

/Project Architecture/Serial Interface/

Company:

Nexvantis

Board Name:

DAQ Core

File Name:

Serial Interface.kicad sch

RS232

Variant:

DRAFT

Git Hash:

41b59e0

Project Name:

Data Acquisition Board

Date:

2025-0

Revision:

+ (Unreleased)

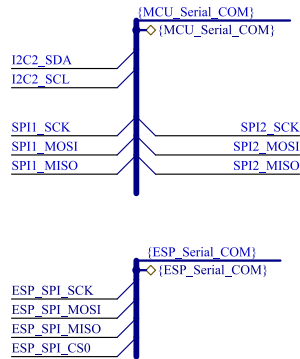
Size:

Sheet:

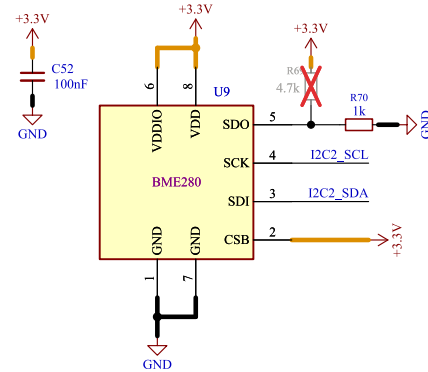
9 of 13

[10] Sensor

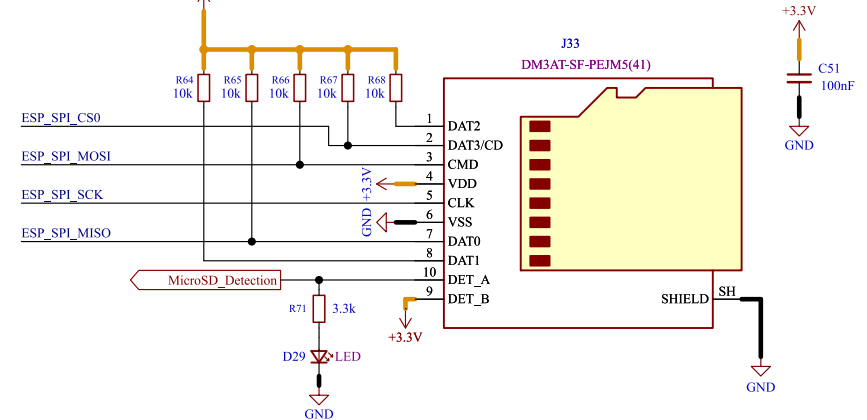
BUS



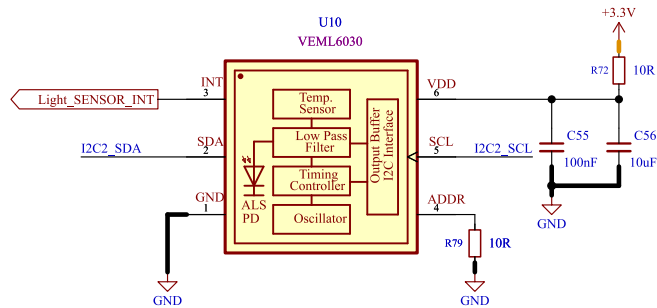
BME280 : Temperature, humidity and Pressure sensor



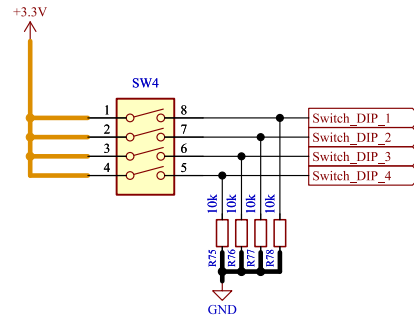
Micro SD Card



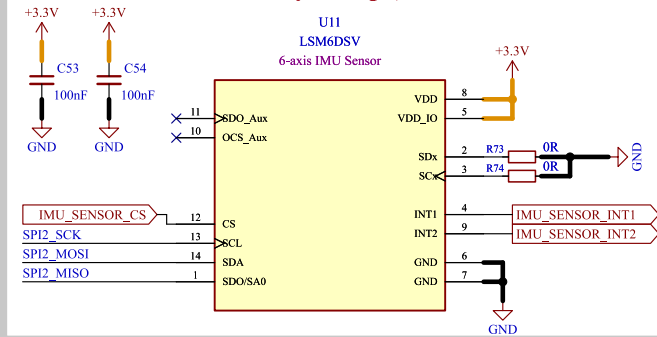
VEML6030 : Light sensor



Switch DIP



BME280 : IMU Sensor (Accelerometer and Gyroscope)



Comments:

Company:

Nexvantis

Variant:

DRAFT

Git Hash:

41b59e0

Board Name:

DAQ Core

Project Name:

Data Acquisition Board

Sheet Title:

Sensor

File Name:

Sensor.kicad_sch

Designer:

Jordan Nzokou

Date:

2025-01-12

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Sensor/

Reviewer:

Size:

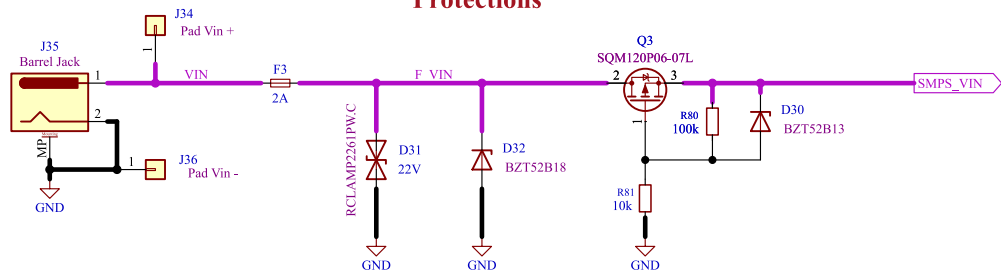
A4

Sheet:

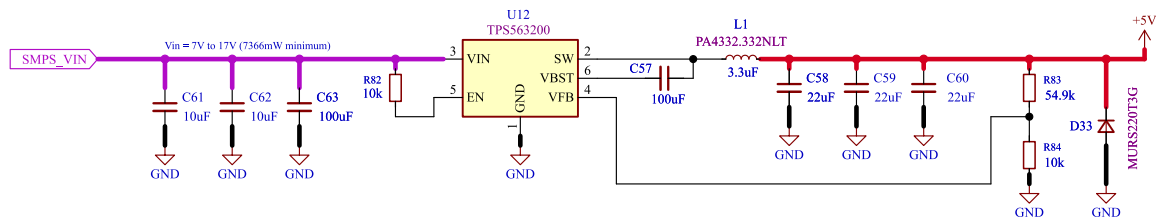
10 of 13

[11] Power

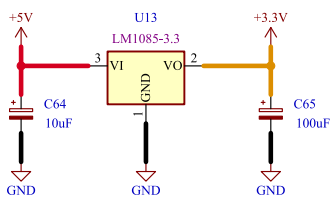
Protections



5V SMPS Regulator



3V3 LDO Regulator



Protections:
->Overvoltage
->Voltage spike
->Overcurrent
->Reverse Polarity

Comments:
LM1085-3.3 Datasheet
TPS563200 Datasheet

Company:
Nexvantis

Board Name:
DAQ Core

File Name:
Power.kicad_sch

Reviewer:

Variant:
DRAFT

Project Name:
Data Acquisition Board

Date:
2025-01-12

Size:
A4

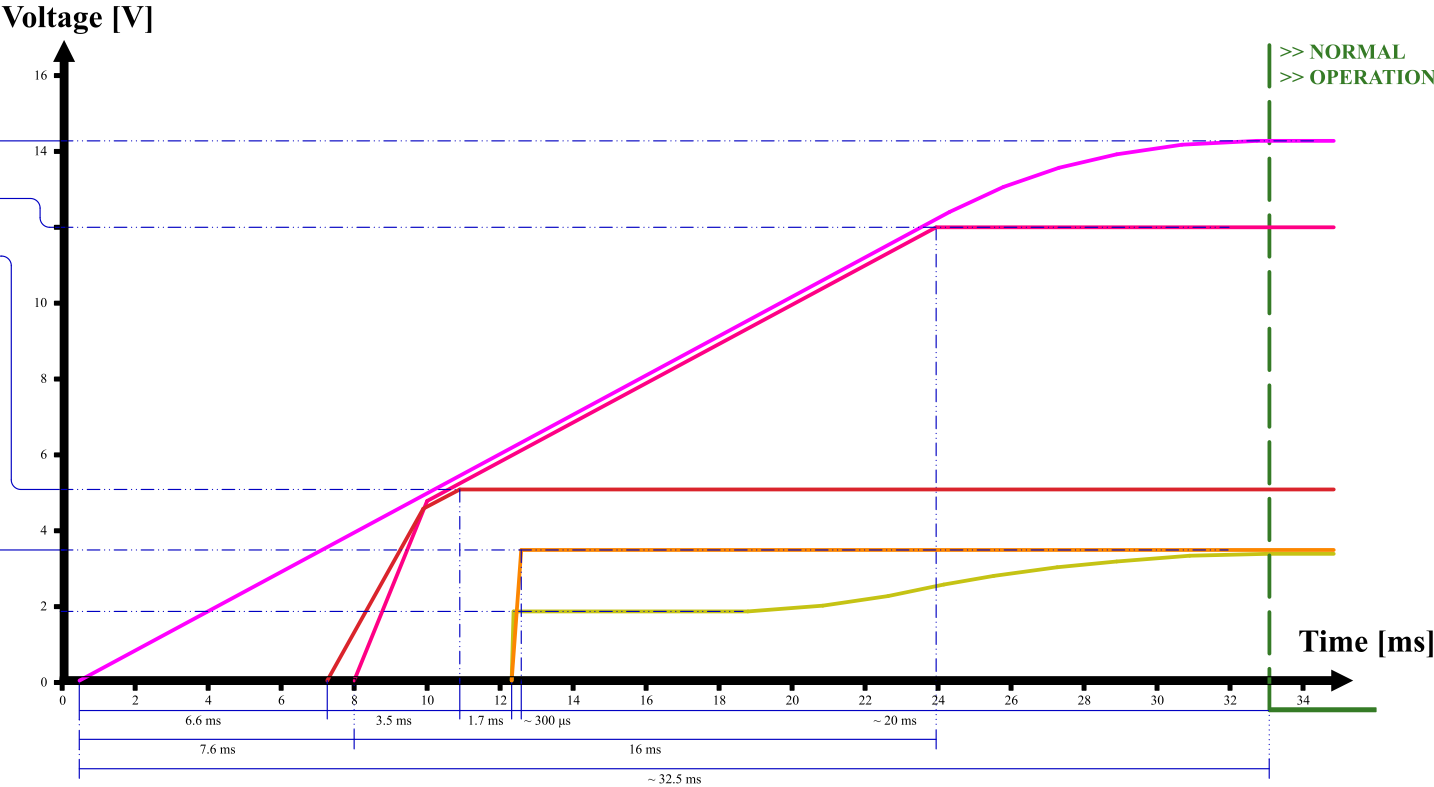
Git Hash:
41b59e0

Revision:
+ (Unreleased)

Sheet:
11 of 13

[12] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



	Comments:	Company: Nexvantis	Variant: DRAFT	Git Hash: 41b59e0
		Board Name: DAQ Core	Project Name: Data Acquisition Board	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12
	Sheet Path: /Power - Sequencing/	Reviewer:	Size: A4	Revision: + (Unreleased)
				Sheet: 12 of 13

[13] Revision History

	Comments:	Company: Nexvantis		Variant: DRAFT	Git Hash: 41b59e0
		Board Name: DAQ Core		Project Name: Data Acquisition Board	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Jordan Nzokou	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 13 of 13